

CSE 309

Tuesday, Jan 30, 2024



1. a for some $a \in \Sigma$

$$\Sigma = \{0, 1\}$$

$$\begin{array}{c} \underline{0} \\ 1 \end{array} \text{ is } \boxed{\begin{array}{l} \text{R.E. r.e.} \\ \text{re RE} \end{array}}$$

2. ϵ

3. ϕ

null set // empty set

$$\Sigma = \{a, b, c\} = \underline{\{a\}} \cup \underline{\{b\}} \cup \underline{\{c\}}$$

$$1+2=3$$

$$\begin{aligned} & (a \cup b)^* \\ &= \{ \epsilon, a, b, aa, aaa, \dots \\ & \quad bh, bhh, \dots \\ & \quad ab, ba, aab, bab, \dots \} \end{aligned}$$

$$\Sigma = \{0, 1\}$$

// assume we have single digit alphabets.

$$0^*10^* = \{w \mid w \text{ contains one } 1\}$$

$$\Sigma^*1\Sigma^* = \{w \mid w \text{ has at least one } 1\}$$

$$\boxed{(\Sigma \circ \Sigma)^*} = \{w \mid w \text{ is a string of even length}\}$$

$$\Sigma \Sigma$$

$$(01)^*$$

$$\Sigma = \{0, 1, 1\}$$

$$\Sigma = \{0, 1, 1\}$$

$a \quad b$

$$a=0, b=1$$

$$(0 \cup \varepsilon)^* = 0^* \cup 1^*$$

$$1^* \phi = \phi$$

$$1^* \varepsilon = 1^*$$

$$\phi^* = \{\varepsilon\}$$

$$R \cup \phi = R$$

$$R \circ \varepsilon = R$$

$$R \cup \varepsilon \stackrel{?}{=} R$$

$$\frac{R=\emptyset}{R \cup \varepsilon} = \underline{\{\emptyset, \varepsilon\}} \neq R$$

$$R \circ \phi = R$$

$$\phi \cup \phi = \phi$$

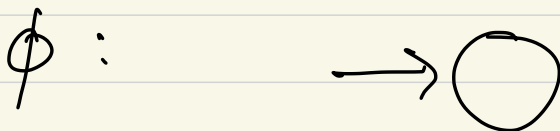
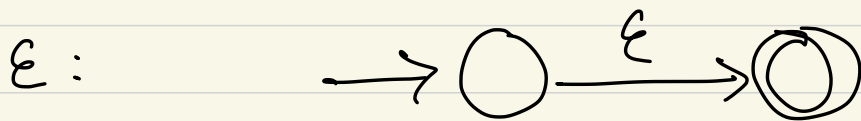
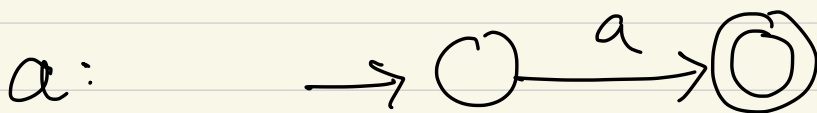
Theorem:

A language is regular
if and only if

Some regular expression
describes it.

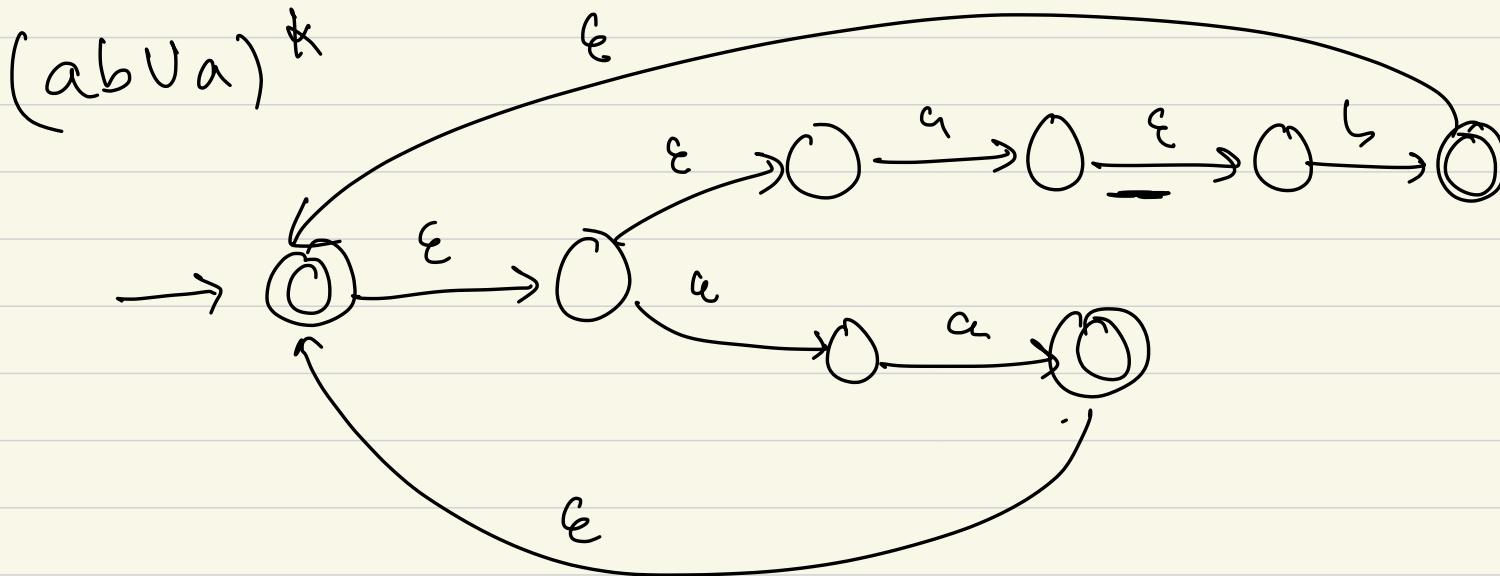
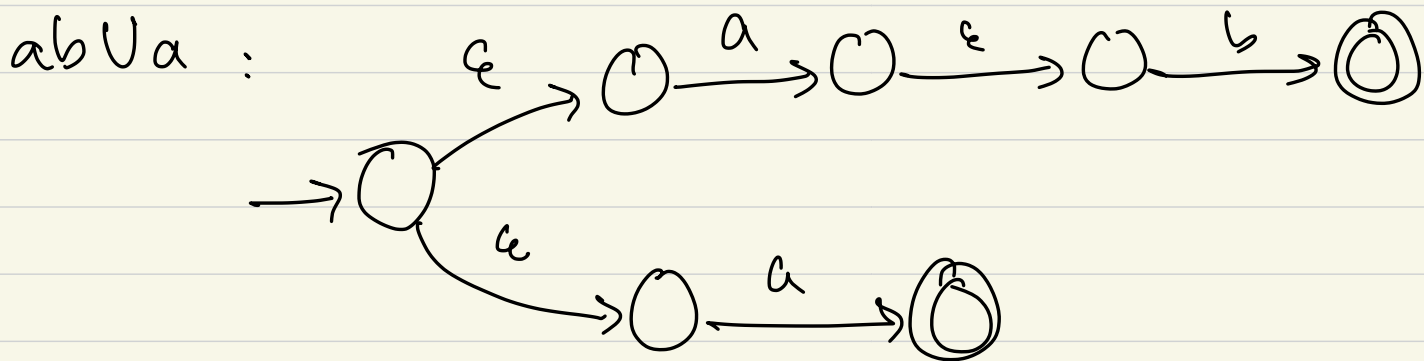
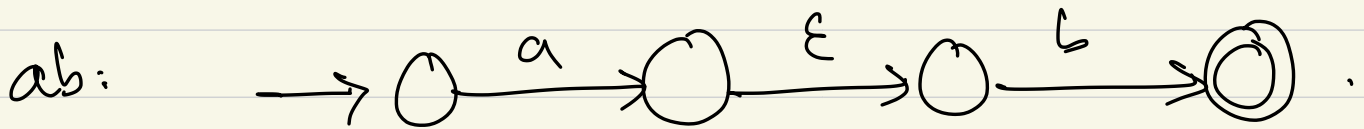
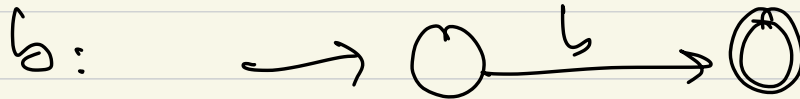
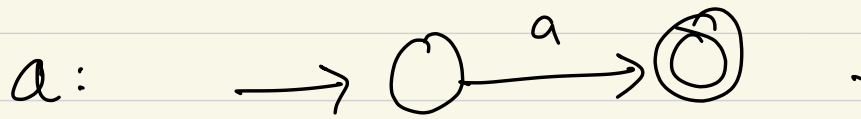
Proof: $a \in \Sigma$, ϵ , ϕ , $\cup$, $\circ$, $*$
atomic re operators

$a \in \Sigma$



$$\underline{\Sigma = \{a, b\}}$$

$$, \quad \underline{(ab \cup a)^*}$$



Example: $\Sigma = \{0, 1\}$

$$L = (01 \cup 1)^* 11$$

NFA N for some
language A .

Can we construct an NFA N'
for A^* ?

